Week 4 complete solutions

Problem-sheet solutions

Each solution states the intended method and relevant conditions, shows the marked reasoning, gives a final result where useful, and identifies specific common mistakes. Award method credit after a purely arithmetic slip when the setup remains valid.

Question 1 — partial derivatives and a Hessian

`W04-Q01-PA (M6)` **Method and conditions.** Work on the stated domain $x > 0$, which makes $\log x$ and its derivatives well-defined. Hold the other variable fixed for each partial derivative.

The first partial derivatives are

$$f_x(x, y) = 2xe^y + \frac{y}{x},$$
$$f_y(x, y) = x^2e^y + \log x.$$

Therefore the column gradient is

$$\nabla f(x, y) = \begin{bmatrix} 2xe^y + y/x \\ x^2e^y + \log x \end{bmatrix}.$$

At $(1, 0)$,

$$\nabla f(1, 0) = \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

Marks (5). Correct f_x, 1; correct f_y, 1; column orientation and input order, 1; two evaluated entries, 2.

Common mistakes. Omitting the chain derivative in e^y; treating $y \log x$ as constant when differentiating with respect to x; writing the gradient as a row.

`W04-Q01-PB (M6)` **Method and conditions.** Differentiate the first partials from Part A again. All second partial derivatives are continuous on $(0, \infty) \times \mathbb{R}$, so the mixed partials agree there.

The second partial derivatives are

$$f_{xx}(x, y) = 2e^y - \frac{y}{x^2},$$
$$f_{xy}(x, y) = 2xe^y + \frac{1}{x},$$
$$f_{yx}(x, y) = 2xe^y + \frac{1}{x},$$
$$f_{yy}(x, y) = x^2e^y.$$

Thus,

$$\mathbf{H}_f(x, y) = \begin{bmatrix} 2e^y - y/x^2 & 2xe^y + 1/x \\ 2xe^y + 1/x & x^2e^y \end{bmatrix},$$

and

$$\boxed{\mathbf{H}_f(1,0) = \begin{bmatrix} 2 & 3 \\ 3 & 1 \end{bmatrix}.}$$

The off-diagonal entries agree because f_{xy} and f_{yx} are continuous and equal throughout the domain.

Marks (6). Four correct second partial derivatives, 3; correct Hessian at the point, 2; continuity condition and mixed-partial conclusion, 1.

Common mistakes. Using a positive sign in the derivative of y/x; placing a first derivative in the Hessian; asserting mixed-partial symmetry without checking the domain and continuity condition.

Question 2 — a gradient and local change

`W04-Q02-PA (M6)` **Method and conditions.** Differentiate the exponential by the scalar chain rule and list derivatives in the input order x, y, z.

$$\begin{aligned} g_x(x,y,z) &= y^2 - e^{z-x}, \\ g_y(x,y,z) &= 2xy, \\ g_z(x,y,z) &= e^{z-x}. \end{aligned}$$

Therefore,

$$\nabla g(x,y,z) = \begin{bmatrix} y^2 - e^{z-x} \\ 2xy \\ e^{z-x} \end{bmatrix}.$$

At $\mathbf{a} = [1,2,1]^{\mathsf{T}}$, the exponent is zero, so

$$\boxed{\nabla g(\mathbf{a}) = \begin{bmatrix} 3 \\ 4 \\ 1 \end{bmatrix}.}$$

Marks (6). Three correct first partial derivatives, 3; column orientation and order, 1; correct evaluation, 2.

Common mistakes. Missing the minus sign in g_x; differentiating xy^2 with respect to y as xy; evaluating e^{z-x} as e.

`W04-Q02-PB (M6)` **Method and conditions.** For a differentiable scalar-valued function, the first-order change is the total differential $\mathrm{d}g = \nabla g(\mathbf{a})^{\mathsf{T}} \mathrm{d}\mathbf{x}$.

The dimensions and calculation are

$$\begin{aligned} \mathrm{d}g &= \underbrace{\begin{bmatrix} 3 & 4 & 1 \end{bmatrix}}_{1\times 3} \underbrace{\begin{bmatrix} 0.02 \\ -0.01 \\ 0.03 \end{bmatrix}}_{3\times 1} \\ &= 3(0.02) + 4(-0.01) + 1(0.03) \\ &= 0.05. \end{aligned}$$

Final result and interpretation. The first-order approximation predicts a scalar increase of 0.05. The result is scalar because the dimension of the product is $(1\times 3)(3\times 1) = 1\times 1$. It is the linear part of the local change. For the stated finite displacement, omitted higher-order terms mean that it need not equal $g(\mathbf{a}+\mathrm{d}\mathbf{x}) - g(\mathbf{a})$ exactly.

Marks (6). Correct differential setup, 1; correct dimensions, 1; three-term dot product and value, 2; scalar explanation, 1; approximation qualification, 1.

Common mistakes. Multiplying the two columns entry by entry without summing; omitting the transpose; calling the differential an exact finite change.

Question 3 — a directional second derivative

`W04-Q03-PA (M6)` **Method and conditions.** Form the Hessian, evaluate it, and use the quadratic expression for the second derivative along the straight path. The function is a polynomial, so all required derivatives are continuous.

The first partials are

$$f_x = 3x^2 + 2y^2 - 6y, \qquad f_y = 4xy - 6x.$$

Hence,

$$\mathbf{H}_f(x,y) = \begin{bmatrix} 6x & 4y-6 \\ 4y-6 & 4x \end{bmatrix}, \qquad \mathbf{H}_f(1,-1) = \begin{bmatrix} 6 & -10 \\ -10 & 4 \end{bmatrix}.$$

For $\mathbf{v} = [1,2]^\mathsf{T}$,

$$\mathbf{H}_f(1,-1)\mathbf{v} = \begin{bmatrix} -14 \\ -2 \end{bmatrix},$$

so

$$\mathbf{v}^\mathsf{T}\mathbf{H}_f(1,-1)\mathbf{v} = \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} -14 \\ -2 \end{bmatrix} = -18.$$

For a direct check, substitution into the path gives

$$\begin{aligned} \varphi(t) &= (1+t)^3 + 2(1+t)(-1+2t)^2 - 6(1+t)(-1+2t) \\ &= 9 - 9t - 9t^2 + 9t^3. \end{aligned}$$

Therefore,

$$\varphi''(t) = -18 + 54t, \qquad \boxed{\varphi''(0) = -18.}$$

This agrees with the Hessian calculation. No global or local curvature classification follows from this one directional value.

Marks (7). Correct general Hessian, 2; correct evaluated Hessian, 1; matrix–vector and quadratic calculation, 2; direct path expansion or differentiation and agreement, 2.

Common mistakes. Using the gradient instead of the Hessian; omitting one cross term in the quadratic expression; following the direction $[1,1]^\mathsf{T}$ instead of $[1,2]^\mathsf{T}$; inferring a global curvature classification from one direction at one point.

Question 4 — scalar chain rules

`W04-Q04-PA (M6)` **Method and conditions.** Use the scalar-path chain rule with a column gradient and column path derivative.

$$\nabla f(x,y) = \begin{bmatrix} 2xy \\ x^2 \end{bmatrix}, \qquad \mathbf{g}'(t) = \begin{bmatrix} 1 \\ e^t \end{bmatrix}.$$

At $t = 0$,

$$\mathbf{g}(0) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad \nabla f(\mathbf{g}(0)) = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \qquad \mathbf{g}'(0) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

The derivative is therefore

$$\frac{\mathrm{d}}{\mathrm{d}t} f(\mathbf{g}(t))\Big|_{t=0} = \underbrace{\begin{bmatrix} 2 & 1 \end{bmatrix}}_{1\times 2} \underbrace{\begin{bmatrix} 1 \\ 1 \end{bmatrix}}_{2\times 1} = \boxed{3}.$$

Marks (5). Correct gradient, 1; correct path derivative, 1; evaluation of both at $t = 0$, 1; dimensional setup, 1; final derivative, 1.

Common mistakes. Evaluating the gradient at $(0, 0)$ rather than at $\mathbf{g}(0)$; omitting the derivative of e^t; leaving the product as a two-entry vector.

`W04-Q04-PB (M6)` **Method and conditions.** Since the outer function is scalar-valued, use

$$\nabla h = \mathbf{J}_{\mathbf{G}}^{\mathsf{T}} \nabla F(\mathbf{G}).$$

The condition $v > 0$ holds because the second component of $\mathbf{G}$ is $e^y > 0$.

The outer gradient and inner Jacobian are

$$\nabla F(u,v) = \begin{bmatrix} v \\ u + 1/v \end{bmatrix} \quad \text{and} \quad \mathbf{J}_{\mathbf{G}}(x,y) = \begin{bmatrix} 2x & 1 \\ 0 & e^y \end{bmatrix}.$$

At $(1, 0)$,

$$\mathbf{G}(1,0) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \nabla F(\mathbf{G}(1,0)) = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \mathbf{J}_{\mathbf{G}}(1,0) = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}.$$

Thus,

$$\nabla h(1,0) = \underbrace{\begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}}_{2\times 2} \underbrace{\begin{bmatrix} 1 \\ 2 \end{bmatrix}}_{2\times 1} = \boxed{\begin{bmatrix} 2 \\ 3 \end{bmatrix}}.$$

For verification,

$$h(x,y) = (x^2 + y)e^y + y,$$

because $\log(e^y) = y$. Direct differentiation gives

$$h_x = 2xe^y$$

and

$$h_y = e^y + (x^2 + y)e^y + 1.$$

At $(1, 0)$, these entries are 2 and 3, agreeing with the chain rule.

Marks (7). Correct outer gradient and intermediate point, 2; correct inner Jacobian, 1; dimensionally correct chain-rule product, 1; correct column result, 1; explicit composition and direct verification, 2.

Common mistakes. Using $\mathbf{J_G}\nabla F$ without transposing the inner Jacobian; evaluating the outer gradient at $(1, 0)$ instead of $\mathbf{G}(1, 0)$; using $\log(e^y) = e^{\log y}$.

Question 5 — a vector-valued derivative

`W04-Q05-PA (M6)` **Method and conditions.** Put the partial derivatives of each output in one row. There are two outputs and three inputs, so the Jacobian is 2×3.

$$\mathbf{J_F}(x, y, z) = \begin{bmatrix} y & x & 2z \\ e^x & z & y \end{bmatrix}.$$

At $\mathbf{a} = [0, 1, 2]^\mathsf{T}$,

$$\boxed{\mathbf{J_F}(\mathbf{a}) = \begin{bmatrix} 1 & 0 & 4 \\ 1 & 2 & 1 \end{bmatrix}.}$$

Marks (6). Correct first row, 2; correct second row, 2; correct 2×3 dimension, 1; correct evaluation, 1.

Common mistakes. Writing component gradients as columns and obtaining a 3×2 array; using z^2's derivative as z; differentiating e^x as xe^{x-1}.

`W04-Q05-PB (M6)` **Method and conditions.** Apply the linear map represented by the 2×3 Jacobian to the 3×1 input displacement.

$$\begin{aligned} \mathrm{d}\mathbf{F} &= \underbrace{\begin{bmatrix} 1 & 0 & 4 \\ 1 & 2 & 1 \end{bmatrix}}_{2\times 3} \underbrace{\begin{bmatrix} 0.02 \\ -0.01 \\ 0.01 \end{bmatrix}}_{3\times 1} \\ &= \begin{bmatrix} 0.02 + 0 + 0.04 \\ 0.02 - 0.02 + 0.01 \end{bmatrix} \\ &= \boxed{\begin{bmatrix} 0.06 \\ 0.01 \end{bmatrix}}. \end{aligned}$$

The result is 2×1, one first-order change per output. Its first entry approximates the change in $F_1 = xy + z^2$, and its second entry approximates the change in $F_2 = e^x + yz$.

Marks (7). Dimensionally correct product, 2; correct first output change, 2; correct second output change, 2; component interpretation, 1.

Common mistakes. Reversing the multiplication order; returning three changes because there are three inputs; assigning the first row's result to the second output.

Question 6 — composition of vector-valued maps

`W04-Q06-PA (M6)` **Method and conditions.** Evaluate the intermediate point and multiply the outer and inner Jacobians in chain-rule order.

$$\mathbf{g}(1,2) = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}.$$

The inner Jacobian is

$$\mathbf{J}_{\mathbf{g}}(s,t) = \begin{bmatrix} 1 & 1 \\ t & s \\ 2s & 0 \end{bmatrix},$$

so

$$\mathbf{J}_{\mathbf{g}}(1,2) = \underbrace{\begin{bmatrix} 1 & 1 \\ 2 & 1 \\ 2 & 0 \end{bmatrix}}_{3\times 2}.$$

The outer Jacobian is

$$\mathbf{J}_{\mathbf{F}}(x,y,z) = \begin{bmatrix} 1 & 1 & 1 \\ y & x & 0 \end{bmatrix},$$

and hence

$$\mathbf{J}_{\mathbf{F}}(\mathbf{g}(1,2)) = \underbrace{\begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 0 \end{bmatrix}}_{2\times 3}.$$

The chain rule gives

$$\begin{aligned} \mathbf{J}_{\mathbf{H}}(1,2) &= \mathbf{J}_{\mathbf{F}}(\mathbf{g}(1,2))\mathbf{J}_{\mathbf{g}}(1,2) \\ &= \begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 2 & 1 \\ 2 & 0 \end{bmatrix} \\ &= \boxed{\begin{bmatrix} 5 & 2 \\ 8 & 5 \end{bmatrix}}. \end{aligned}$$

The dimensions reconcile as $(2 \times 3)(3 \times 2) = 2 \times 2$.

Marks (7). Correct intermediate point, 1; correct inner Jacobian and evaluation, 2; correct outer Jacobian and evaluation, 2; correct chain-rule product and result, 2.

Common mistakes. Evaluating the outer Jacobian at $(1,2)$; giving the inner Jacobian only two rows; multiplying the factors in reverse order.

`W04-Q06-PB (M6)` **Method and conditions.** Compose first, then differentiate each explicit output with respect to the two original inputs.

The components are

$$H_1(s,t) = s + t + st + s^2,$$
$$H_2(s,t) = (s+t)(st) = s^2t + st^2.$$

Therefore,

$$\mathbf{J_H}(s,t) = \begin{bmatrix} 1+t+2s & 1+s \\ 2st+t^2 & s^2+2st \end{bmatrix}.$$

At $(1,2)$,

$$\boxed{\mathbf{J_H}(1,2) = \begin{bmatrix} 5 & 2 \\ 8 & 5 \end{bmatrix},}$$

which agrees with Part A.

The reversed product would have dimensions

$$\underbrace{\mathbf{J_g}}_{3\times 2}\,\underbrace{\mathbf{J_F}}_{2\times 3},$$

so it is a defined 3×3 matrix. However, $\mathbf{H} : \mathbb{R}^2 \to \mathbb{R}^2$, whose Jacobian must be 2×2. The reversed product therefore cannot be $\mathbf{J_H}$.

Marks (8). Two correct explicit components, 2; correct general direct Jacobian, 3; correct evaluation and reconciliation, 1; dimension-based explanation of the reversed product, 2.

Common mistakes. Omitting st from the first component; applying the product rule incompletely to $(s+t)st$; claiming that the reversed product is undefined instead of explaining that it has the wrong dimension.

Question 7 — marginal effects in a production index

`W04-Q07-PA (M6)` **Method and conditions.** The positivity conditions $K > 0$ and $L > 0$ justify all displayed powers. Differentiate while holding the other input fixed, then use the total differential for the local change.

The first partial derivatives are

$$Y_K(K,L) = K^{-1/2}L^{1/2}, \qquad Y_L(K,L) = K^{1/2}L^{-1/2}.$$

Thus,

$$\boxed{\nabla Y(4,9) = \begin{bmatrix} 3/2 \\ 2/3 \end{bmatrix}.}$$

The second partial derivatives are

$$
\begin{aligned}
Y_{KK}(K,L) &= -\frac{1}{2}K^{-3/2}L^{1/2},\\
Y_{KL}(K,L) &= \frac{1}{2}K^{-1/2}L^{-1/2},\\
Y_{LK}(K,L) &= \frac{1}{2}K^{-1/2}L^{-1/2},\\
Y_{LL}(K,L) &= -\frac{1}{2}K^{1/2}L^{-3/2}.
\end{aligned}
$$

Therefore,

$$
\boxed{\mathbf{H}_Y(4,9) = \begin{bmatrix} -3/16 & 1/12 \\ 1/12 & -1/27 \end{bmatrix}.}
$$

The displaced point is

$$
\left(4+\frac{1}{10}, 9-\frac{3}{20}\right) = \left(\frac{41}{10}, \frac{177}{20}\right).
$$

Both coordinates are positive, so this point remains in the stated domain.

The first-order change is

$$
\begin{aligned}
\mathrm{d}Y &= \begin{bmatrix} 3/2 & 2/3 \end{bmatrix} \begin{bmatrix} 1/10 \\ -3/20 \end{bmatrix}\\
&= \frac{3}{20} - \frac{1}{10} = \boxed{\frac{1}{20}}.
\end{aligned}
$$

The partial derivative $Y_K(4,9) = 3/2$ is the local rate of change in the index with respect to capital while labour is held fixed. Similarly, $Y_L(4,9) = 2/3$ holds capital fixed. Combining them gives a local linear approximation for simultaneous changes. A finite change also contains higher-order terms, so 1/20 is not asserted to be exact.

Marks (8). Correct gradient at the point, 2; correct Hessian at the point, 3; correct differential calculation, 2; interpretation and approximation limitation, 1.

Common mistakes. Losing the factor 1/2 in second derivatives; evaluating $9^{-3/2}$ as 1/9; interpreting a partial derivative without holding the other input fixed; calling the differential exact.

Question 8 — a Hessian through a scalar composition

`W04-Q08-PA (M6)` **Method and conditions.** Differentiate the gradient product $\phi'(g)\nabla g$ componentwise. Continuous second derivatives justify the Hessians and their symmetry.

For $i, j \in \{1, 2\}$,

$$
\begin{aligned}
\frac{\partial^2 h}{\partial x_j \partial x_i} &= \frac{\partial}{\partial x_j}\left(\phi'(g)\frac{\partial g}{\partial x_i}\right)\\
&= \phi''(g)\frac{\partial g}{\partial x_j}\frac{\partial g}{\partial x_i} + \phi'(g)\frac{\partial^2 g}{\partial x_j \partial x_i}.
\end{aligned}
$$

Collecting the entries gives

$$
\boxed{\mathbf{H}_h = \phi''(g)\nabla g \nabla g^{\mathsf{T}} + \phi'(g)\mathbf{H}_g.}
$$

Here ∇g is 2×1, so $\nabla g \nabla g^{\mathsf{T}}$ is $(2 \times 1)(1 \times 2) = 2 \times 2$. Both Hessians and therefore both terms on the right are 2×2.

For the application, $g(x, y) = x^2 + 2y^2 \geq 0$, so its values lie in the stated interval $I = (-1, \infty)$, on which the outer logarithmic function is defined. Take

$$g(x, y) = x^2 + 2y^2, \qquad \phi(s) = \log(1 + s).$$

At $(1, 1)$,

$$g = 3, \qquad \nabla g = \begin{bmatrix} 2 \\ 4 \end{bmatrix}, \qquad \mathbf{H}_g = \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix}.$$

Also,

$$\phi'(3) = \frac{1}{4}, \qquad \phi''(3) = -\frac{1}{16}.$$

The outer product is

$$\nabla g \nabla g^{\mathsf{T}} = \begin{bmatrix} 4 & 8 \\ 8 & 16 \end{bmatrix}.$$

Therefore,

$$\begin{aligned} \mathbf{H}_h(1, 1) &= -\frac{1}{16} \begin{bmatrix} 4 & 8 \\ 8 & 16 \end{bmatrix} + \frac{1}{4} \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix} \\ &= \begin{bmatrix} -1/4 & -1/2 \\ -1/2 & -1 \end{bmatrix} + \begin{bmatrix} 1/2 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \boxed{\begin{bmatrix} 1/4 & -1/2 \\ -1/2 & 0 \end{bmatrix}}. \end{aligned}$$

For a direct off-diagonal check, let $D = 1 + x^2 + 2y^2$. Since

$$h_x = \frac{2x}{D},$$

we obtain

$$h_{xy} = -\frac{2x(4y)}{D^2} = -\frac{8xy}{D^2}.$$

At $(1, 1)$, $D = 4$, so $h_{xy} = -8/16 = -1/2$, agreeing with the formula.

Marks (10). Correct componentwise product/chain-rule derivation, 3; dimension explanation, 1; correct values of g, its derivatives and the outer derivatives, 2; correct matrix assembly and result, 3; direct off-diagonal check, 1.

Common mistakes. Replacing the outer product with an inner product; omitting the $\phi''(g)$ term; using $\phi''(s) = -1/(1 + s)$; forgetting the factor $4y$ in the direct mixed partial.

Question 9 — Mathematics retrieval

`W04-Q09-PA (M3 retrieval)` **Method and conditions.** Use the value and first two derivatives of $f(x) = e^x$ at the centre 0.

$$f(0) = f'(0) = f''(0) = 1.$$

Thus the second-order Taylor approximation about 0 is

$$T_2(x) = 1 + x + \frac{x^2}{2}.$$

At $x = 0.1$,

$$\boxed{e^{0.1} \approx 1 + 0.1 + \frac{(0.1)^2}{2} = 1.105.}$$

Marks (3). Correct second-order polynomial, 2; correct approximate value with approximation notation, 1.

Common mistakes. Omitting the factor 1/2; centring at $x = 1$; writing equality to the truncated polynomial.

`W04-Q09-PB (M4 retrieval)` **Method and conditions.** The matrix is 2×3 and the vector is 3×1, so the product exists and is 2×1.

$$\begin{aligned}\mathbf{Ab} &= \begin{bmatrix}1 & 0 & -1\\ 2 & 1 & 0\end{bmatrix}\begin{bmatrix}2\\ -1\\ 3\end{bmatrix}\\ &= \begin{bmatrix}2+0-3\\ 4-1+0\end{bmatrix}\\ &= \boxed{\begin{bmatrix}-1\\ 3\end{bmatrix}}.\end{aligned}$$

Marks (3). Correct three dimensions, 1; correct row–column products, 1; correct column result, 1.

Common mistakes. Treating the product as entrywise multiplication; reporting a 3×1 result; losing the negative contribution from the third entry.

Question 10 — Statistics consolidation

`W04-Q10-PA (S3 retrieval)` **Method and conditions.** Under the rate parameterisation, $X \sim$ Exponential(2) has support $x \geq 0$ and survival probability $\mathbb{P}(X > x) = e^{-2x}$ for $x \geq 0$.

The event endpoints lie in the support, and continuity means endpoint inclusion has no effect on their individual probabilities. Hence,

$$\begin{aligned}\mathbb{P}\left(\frac{1}{2} < X \leq 1\right) &= \mathbb{P}\left(X > \frac{1}{2}\right) - \mathbb{P}(X > 1)\\ &= \boxed{e^{-1} - e^{-2}}.\end{aligned}$$

Marks (3). Correct support and rate interpretation, 1; valid interval or CDF setup, 1; correct exact probability, 1.

Common mistakes. Treating 2 as the mean; using support $\{0, 1, ...\}$; reversing the subtraction of the two survival probabilities.

`W04-Q10-PB (S4 retrieval)` **Method and conditions.** Apply linearity of expectation and the variance-of-a-linear-combination identity, retaining the covariance term.

$$\mathbb{E}[Z] = 2\mathbb{E}[X] - \mathbb{E}[Y] = 2(1) - 2 = \boxed{0}.$$

For the variance,

$$\begin{aligned}\operatorname{Var}(Z) &= \operatorname{Var}(2X - Y) \\ &= 2^2 \operatorname{Var}(X) + (-1)^2 \operatorname{Var}(Y) + 2(2)(-1) \operatorname{Cov}(X, Y) \\ &= 4(2) + 3 - 4(1) \\ &= \boxed{7}.\end{aligned}$$

The covariance term is negative because the coefficients of X and Y in Z have opposite signs: $2ab = 2(2)(-1) = -4$.

Marks (3). Correct expectation, 1; correct variance including covariance and final value, 1; correct sign explanation, 1.

Common mistakes. Squaring the expectations rather than the coefficients; omitting the covariance term; using a positive cross term despite the negative coefficient of Y.

Self-test solutions and error check

Solutions and error check

Solution to `W04-ST01-PA (M6)` **Method.** Differentiate once for the column gradient and again for the Hessian.

$$\nabla f(x, y) = \begin{bmatrix} 2x + y \\ x + 2y \end{bmatrix},$$

so

$$\boxed{\nabla f(1, -1) = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.}$$

The second partial derivatives are constant:

$$\boxed{\mathbf{H}_f(1, -1) = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.}$$

Marks (5). Two correct first partials, 2; column orientation and evaluation, 1; correct Hessian diagonal, 1; correct mixed entries, 1.

Common mistakes. Writing the gradient as a row; omitting the mixed derivative of xy; substituting the point before differentiating.

Solution to `W04-ST02-PA (M6)` **Method.** A map with two inputs and three outputs has a 3×2 Jacobian, with one output component per row.

$$\mathbf{J_F}(x,y) = \begin{bmatrix} 1 & 1 \\ y & x \\ 2x & 0 \end{bmatrix}.$$

Therefore,

$$\boxed{\mathbf{J_F}(1,-1) = \begin{bmatrix} 1 & 1 \\ -1 & 1 \\ 2 & 0 \end{bmatrix}.}$$

Marks (5). Correct dimension, 1; correct first row, 1; correct second row, 1; correct third row, 1; correct evaluation, 1.

Common mistakes. Reversing the 3×2 dimension; putting component gradients in columns; differentiating x^2 with respect to y as $2x$.

Solution to `W04-ST03-PA (M6)` **Method.** The condition $x > 0$ holds near $t = 0$ because the first path component is $1 + t$. Use $\nabla F(\mathbf{g}(t))^{\mathsf{T}} \mathbf{g}'(t)$.

$$\nabla F(x,y) = \begin{bmatrix} y + 1/x \\ x \end{bmatrix}, \qquad \mathbf{g}'(t) = \begin{bmatrix} 1 \\ -2 \end{bmatrix}.$$

At $t = 0$,

$$\mathbf{g}(0) = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \text{and} \quad \nabla F(\mathbf{g}(0)) = \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

Thus,

$$\frac{\mathrm{d}}{\mathrm{d}t} F(\mathbf{g}(t))\Big|_{t=0} = \underbrace{\begin{bmatrix} 2 & 1 \end{bmatrix}}_{1\times 2} \underbrace{\begin{bmatrix} 1 \\ -2 \end{bmatrix}}_{2\times 1} = \boxed{0}.$$

Marks (5). Correct gradient, 1; correct path derivative, 1; correct evaluation point, 1; dimensional setup, 1; final derivative, 1.

Common mistakes. Evaluating the gradient at $(0, 0)$; omitting the derivative of $\log x$; treating the zero result as evidence that both component effects are zero.

Solution to `W04-ST04-PA (S4)` **Method.** Apply the linear-transformation rules for expectation and variance.

$$\boxed{\mathbb{E}[Y] = 1 - 2\mathbb{E}[X] = 1 - 6 = -5.}$$

The constant shift does not affect dispersion, and the multiplier is squared:

$$\boxed{\mathrm{Var}(Y) = (-2)^2 \,\mathrm{Var}(X) = 4(2) = 8.}$$

Adding a constant shifts every realised value by the same amount, so it shifts the expectation but leaves deviations from the expectation unchanged.

Marks (5). Correct expectation setup and value, 2; correct variance setup and value, 2; valid explanation of the constant shift, 1.

Common mistakes. Using $1 - 2\,\mathrm{Var}(X)$; multiplying the variance by -2 rather than 4; claiming that a location shift changes dispersion.

Record the first cause of each error under `M6` or `S4`, then reattempt that part from a blank page in two or three days.